

Knowledge Transfer: `transformer_pruning`

Audience: a developer/researcher picking up this codebase for the first time — either to run the existing experiments, extend them with a new pruning criterion, or port the approach to a new model family.

This document goes deeper than the [README](#): the README sells the architecture in a paragraph each; this document explains *why* each piece exists, what it depends on, what breaks if you change it, and how to extend it safely.

1. What this project is

This project extends pruning criteria originally developed for CNNs (VGG/ ResNet, in the sibling thesis project `pruning_framework_v4` — "Pruning and Quantization Techniques for Deep Neural Network Acceleration", NITK, July 2025) onto transformer models, specifically BERT via HuggingFace `transformers`.

It explores three independent pruning/redundancy signals:

1. **Max-3 weight saliency** — score each neuron/filter by the sum of its 3 largest-magnitude incoming weights; prune the lowest scorers. This is the CNN criterion from the thesis, generalized to also work on `nn.Linear` (BERT's FFN layers).
2. **Attention-head redundancy** — cosine similarity / Lp-distance between attention heads' flattened query weights, to find near-duplicate heads (exploratory, feeds no automated pruning step yet — output is heatmaps for a human to read).
3. **Co-activation ("twin") redundancy** — a criterion *not* in the thesis: run real sentences through a trained model, record which FFN neurons fire together, and treat neurons with near-identical firing patterns (Jaccard/ IoU ≥ 0.95) as redundant regardless of what their weights look like.

Criteria 1 and 3 both terminate in the same physical operation: shrinking a BERT FFN block's (`intermediate`, `output`) `nn.Linear` pair. That shared endpoint is what the architecture (Section 4) is built around.

`pruning_framework_v4` and this repo share a research idea, not code — different frameworks, different model families, different datasets. Don't go looking for imports between them.

See `../pruning_framework_v4/KNOWLEDGE_TRANSFER.md` for the CNN-side thesis framework: its full pruning-criteria catalog with thesis chapter references, the two bugs found/fixed there (K-means centroid init, distance-scorer double-pruning) with from-scratch pure-Python verification code, and the `config.ini` /Kaggle-validation-run plan.

2. Repository layout

```
transformer_pruning/
├── pruning_transformer/  # the importable package – all reusable logic
│   ├── __init__.py      # public API surface (what `from pruning_transformer import X` exposes)
│   ├── layers.py        # LayerHandle, LayerKind, discover_layers
│   ├── model_adapter.py  # FFNLayerAdapter, AttentionLayerAdapter, BertLayerAdapter
│   ├── scoring.py        # SaliencyScorer, Max3SaliencyScorer, lowest_scoring
│   ├── network_scanner.py # NetworkSaliencyScanner
│   ├── head_analysis.py  # AttentionHeadAnalyzer
│   ├── ffn_surgery.py    # FFNSurgeon
│   ├── selectors.py      # NeuronSelector, SaliencySelector, TwinRedundancySelector
│   ├── pruning_workflow.py # prune_ffn_layer (the orchestration entry point)
│   ├── activation_recording.py # ActivationRecorder
│   └── redundancy.py     # JaccardTwinFinder
├── experiments/         # 10 runnable scripts, one per exploration stage
│   ├── 01_vgg_saliency_demo.py
│   ├── ...
│   └── 10_twin_neuron_pruning_experiment.py
├── docs/
│   └── KNOWLEDGE_TRANSFER.md # this file
├── README.md
├── requirements.txt
└── .gitignore
```

Rule of thumb: `pruning_transformer/` **never imports from** `experiments/`, and every experiment script only imports from `pruning_transformer` (plus third-party libraries). If you find yourself importing an experiment script from inside the package, that logic belongs in the package instead.

3. Environment & running

There is **no GPU/PyTorch on the primary dev machine** for this project (same constraint as `pruning_framework_v4`) — these scripts are meant to be run on Colab/Kaggle, not locally. Don't assume `import torch` will succeed in this dev environment; if you need to sanity-check pure-Python logic locally without a real torch install, stub `torch / torch.nn` (see the git history of this repo's session logs for examples of this pattern used during review).

Setup on a machine/notebook that does have PyTorch:

```
pip install -r requirements.txt
```

Run any experiment directly:

```
python experiments/05_bert_mrpc_full_experiment.py
```

Status caveat (from the README): all modules were extracted/cleaned from an exploratory Colab notebook and have not yet been run end-to-end in *this* repo's layout — treat printed results as needing a real run before citing them anywhere. One correctness fix was made during extraction: the original notebook's twin-finder computed `intersection / (fires_A + fires_B)`, which is off by a factor vs. true Jaccard/IOU (`intersection / union`, where `union = fires_A + fires_B - intersection`). `redundancy.py`'s `JaccardTwinFinder` uses the corrected formula — twin-pair counts will differ slightly from the original notebook's printed output.

4. Architecture

The package is built around **four small interfaces** so that adding a new pruning criterion, a new attention metric, a new prunable layer type, or support for a new model family never means editing existing scanning, surgery, or orchestration code.

```
graph TD
    subgraph Interfaces["Abstractions (the extension points)"]
        SS["SaliencyScorer<br/>.score(weight)"]
        NSel["NeuronSelector<br/>.select_keep_indices(weight)"]
        FA["FFNLayerAdapter<br/>.get_ffn / .set_ffn"]
        AA["AttentionLayerAdapter<br/>.num_attention_heads / .hidden_size / .get_query_weight"]
    end

    Max3["Max3SaliencyScorer"] -.implements.-> SS
    SalSel["SaliencySelector"] -.implements.-> NSel
    SalSel -- "wraps a" --> SS
    TwinSel["TwinRedundancySelector"] -.implements.-> NSel
    Bert["BertLayerAdapter"] -.implements.-> FA
    Bert -.implements.-> AA

    Scanner["NetworkSaliencyScanner"] -- "depends on" --> SS
    Workflow["prune_ffn_layer()"] -- "depends on" --> NSel
    Workflow -- "depends on" --> FA
    Workflow -- "uses" --> Surgeon["FFNSurgeon"]
    HeadAn["AttentionHeadAnalyzer"] -- "depends on" --> AA
    Recorder["ActivationRecorder"] -- "depends on" --> FA
    Recorder -- "output feeds" --> Redund["JaccardTwinFinder"]
    Redund -- "twin pairs feed" --> TwinSel
```

The four interfaces

Interface	File	Contract	Implementations	Consumers
SaliencyScorer	scoring.py	score(weight) -> Tensor (one score per output unit)	Max3SaliencyScorer	NetworkSaliencyScanner, SaliencySelector
NeuronSelector	selectors.py	select_keep_indices(intermediate_weight) -> list	SaliencySelector, TwinRedundancySelector	prune_ffn_layer
FFNLayerAdapter	model_adapter.py	get_ffn(layer_idx), set_ffn(layer_idx, intermediate, output)	BertLayerAdapter	prune_ffn_layer, ActivationRecorder
AttentionLayerAdapter	model_adapter.py	num_attention_heads, hidden_size, get_query_weight(layer_idx)	BertLayerAdapter	AttentionHeadAnalyzer

TransformerLayerAdapter (also in model_adapter.py) is just FFNLayerAdapter + AttentionLayerAdapter combined — a convenience type for BertLayerAdapter, which happens to support both. **New consumers should depend on whichever single interface they actually need**, not the combined one — that's the whole point of the split (see Section 4.1).

4.1 Why it's shaped this way (SOLID, with concrete pointers)

This codebase was deliberately refactored (across several review passes) to satisfy SOLID, replacing an earlier version with near-duplicate classes (SaliencyPruner / UniversalPruner, SimilarityScanner / DistributionScanner, a CoActivationScanner that mixed recording with analysis, etc. — see Section 9 for the full old-name → new-location map). Concretely, in this codebase:

- **Single Responsibility** — activation_recording.py only records which neurons fired; redundancy.py only decides which fired-together pairs count as "twins". The original notebook's CoActivationScanner did both, so changing the redundancy threshold logic risked also touching the recording/hooking mechanics. Now they change independently.
- **Open/Closed** — two concrete examples:
 - layers.py's discover_layers recognizes new module types via a LayerKind(name, module_type, name_filter) passed through extra_kinds, not via editing an if/elif chain. Adding nn.LayerNorm support is discover_layers(model, extra_kinds=[LayerKind("layernorm", nn.LayerNorm)]) — zero edits to layers.py.
 - head_analysis.py's compute_distance_matrix looks up the Lp-norm order from a LP_METRICS dict ({"manhattan": 1, "euclidean": 2}) instead of an if metric == "manhattan" else ... ternary. A third metric is a new dict entry; an unrecognized metric raises ValueError instead of silently defaulting to Euclidean.
- **Liskov Substitution** — SaliencySelector and TwinRedundancySelector are fully interchangeable wherever a NeuronSelector is expected; prune_ffn_layer (pruning_workflow.py) never branches on which one it got, and never will need to for a third selector either.
- **Interface Segregation** — model_adapter.py's adapter interface used to be one fat TransformerLayerAdapter with 5 abstract members covering both FFN and attention access. It was split into FFNLayerAdapter and AttentionLayerAdapter because no consumer used both halves (prune_ffn_layer / ActivationRecorder only touch FFN; AttentionHeadAnalyzer only touches attention). A future adapter for a model that only supports FFN-style pruning now implements FFNLayerAdapter alone, without being forced to stub out query-weight/head-count methods it has no use for.
- **Dependency Inversion** — prune_ffn_layer, AttentionHeadAnalyzer, and ActivationRecorder used to reach directly into model.encoder.layer[i].intermediate.dense / ...attention.self.query... — i.e., high-level orchestration depended on one concrete HuggingFace model's attribute layout. They now depend on the adapter interfaces instead; BertLayerAdapter is the only implementation, but a differently shaped architecture (say, a model that nests its FFN differently) is a new adapter class, not an edit to three separate consumers.

4.2 What's *not* abstracted, on purpose

Not every class has an interface, and that's intentional — don't "fix" these without a concrete reason:

- **FFNSurgeon** (ffn_surgery.py) is a concrete class, not an ABC with multiple implementations. There is exactly one correct way to physically resize a (intermediate, output) Linear pair given keep_indices — it's a mechanical operation, not a policy decision, so there's nothing to swap.
- **AttentionHeadAnalyzer** bundles cosine similarity *and* Lp-distance in one class rather than two, because both share the same get_flat_heads extraction step; splitting them would mean either duplicating that extraction or introducing a shared base class for two methods — not worth it for two closely related, rarely-changing computations.
- **BERT-only model coverage**: only BertLayerAdapter exists. The interface split (4.1) makes adding a second adapter (e.g. for a model with a differently-shaped FFN block) straightforward *if and when* one is actually needed — don't pre-build a DistilBertLayerAdapter speculatively.

5. Module-by-module reference

layers.py

- `LayerHandle(name, module, kind)` — a dataclass wrapping a discovered module with its qualified name and a `kind` tag ("conv2d", "linear", or whatever a custom `LayerKind` names it).
- `LayerKind(name, module_type, name_filter=None)` — a frozen dataclass describing one recognizable module type. `name_filter`, if set, requires the module's qualified name to contain that substring (used to isolate BERT's `intermediate` FFN-expansion Linears from all other Linears).
- `discover_layers(model, include_conv=True, include_linear=False, linear_name_filter=None, extra_kinds=())` — walks `model.named_modules()` and returns matching `LayerHandle`s. The built-in conv/linear kinds are gated by `include_conv` / `include_linear`; anything else goes through `extra_kinds`.

model_adapter.py

- `FFNLayerAdapter (ABC)` — `get_ffn(layer_idx) -> (intermediate, output)`, `set_ffn(layer_idx, intermediate, output) -> None`.
- `AttentionLayerAdapter (ABC)` — `num_attention_heads` (property), `hidden_size` (property), `get_query_weight(layer_idx)`.
- `TransformerLayerAdapter` — `FFNLayerAdapter` + `AttentionLayerAdapter`, nothing added on top. Only exists so `BertLayerAdapter` has one type name to inherit that satisfies both.
- `BertLayerAdapter(model)` — the only concrete adapter. Assumes `model.encoder.layer[i]` layout (HuggingFace `BertModel`; also correct for `RobertaModel`, which shares that layout).

scoring.py

- `SaliencyScorer (ABC)` — `score(weight) -> Tensor`, one score per output unit (`weight.shape[0]`).
- `Max3SaliencyScorer` — sums each unit's 3 largest-magnitude incoming weights. Handles both 4D conv weights (`out_ch`, `in_ch`, `kh`, `kw`, summed across `in_ch` and kernel position) and 2D linear weights (`out`, `in`, summed across `in`) via `weight.dim()` branching.
- `lowest_scoring(scorer, weight, amount)` — convenience free function: the `amount` lowest-scoring unit indices + their scores, as `[index, score]` pairs. Used for one-off single-layer lookups (see `experiments/01`).

network_scanner.py

- `NetworkSaliencyScanner(layers, scorer)` — takes a list of `LayerHandle` and a `SaliencyScorer` as constructor dependencies (this is the DIP example: the *same* class handles a pure-CNN pass over VGG16 and a mixed Conv2d+Linear pass over BERT, just by being handed a different `layers` list — see `experiments/02` vs `experiments/03`).
 - `.scan()` → a `pandas.DataFrame` with one row per layer: Layer, Type, Units, Avg/Min/StdDev score.
 - `.weakest_units(layer_name, n=5)` → the `n` lowest-scoring unit indices for one named layer.

head_analysis.py

- `LP_METRICS = {"manhattan": 1, "euclidean": 2}` — module-level dict, the OCP extension point for distance metrics.
- `AttentionHeadAnalyzer(model=None, adapter=None)` — pass either a raw HuggingFace model (wrapped in `BertLayerAdapter` automatically) or a pre-built `AttentionLayerAdapter`. Raises `ValueError` if neither is given.
 - `.get_flat_heads(layer_idx)` → (`num_heads`, `head_dim * hidden_size / num_heads`)-shaped tensor of each head's flattened query weights.
 - `.compute_similarity(layer_idx)` → cosine similarity matrix (heads unit-normalized first).
 - `.compute_distance_matrix(layer_idx, metric="manhattan", normalize=False)` → Lp-distance matrix; `normalize=True` unit-normalizes heads first to isolate directional (vs. magnitude) redundancy. Raises `ValueError` on an unrecognized `metric`.

ffn_surgery.py

- `FFNSurgeon.resize(intermediate, output, keep_indices)` — builds fresh `nn.Linear` modules with only `keep_indices` rows/columns kept, copies weights/biases under `torch.no_grad()`, moves the new modules to the original device, returns (`new_intermediate`, `new_output`). **Does not mutate the model in place** — the caller (`prune_ffn_layer`) is responsible for re-attaching the returned modules via the adapter. **No bounds-checking** on `keep_indices` — passing an out-of-range index will raise from the underlying tensor indexing, not from this function; that's intentional (validation lives with whoever produces the indices, i.e. the selector), but worth knowing if you're debugging an `IndexError` here.

selectors.py

- `NeuronSelector (ABC)` — `select_keep_indices(intermediate_weight) -> list`.
- `SaliencySelector(scorer, prune_percent)` — keeps the top `(100 - prune_percent)%` scoring neurons under the given `SaliencyScorer`.

- `TwinRedundancySelector(twin_pairs)` — drops the higher-indexed neuron of every `(a, b, similarity)` twin tuple. **Ignores the `intermediate_weight` argument's values entirely** (only reads its shape) — this is correct, not a bug: the redundancy signal here is behavioral (from `JaccardTwinFinder`), not weight-based, but the method still accepts the same argument shape as `SaliencySelector` to stay substitutable (LSP).

pruning_workflow.py

- `prune_ffn_layer(model, layer_index, selector, surgeon=None, adapter=None)` — the single orchestration entry point. Defaults `surgeon` to a fresh `FFNSurgeon()` and `adapter` to `BertLayerAdapter(model)` if not given. Returns the number of neurons kept. This is the **only** place a `NeuronSelector` and `FFNSurgeon` meet — if you're adding a new pruning criterion, you should never need to touch this function, only add a new `NeuronSelector`.

activation_recording.py

- `ActivationRecorder(model, tokenizer, adapter=None)` — registers a forward hook on the target layer's FFN intermediate module (via `adapter.get_ffn(layer_idx)[0]`), runs each input text through the model one at a time, records `(output > 0).float()` (ReLU-fired mask) per example, and returns the concatenated tensor. The hook is always removed in a `finally` block, and internal state (`_activations`) is reset after every call — **not thread-safe / not reentrant**, but `record()` can be called repeatedly on the same instance for different layers.

redundancy.py

- `JaccardTwinFinder.find(fires, threshold=0.95)` — takes the tensor `ActivationRecorder.record()` produces (`[num_examples, num_neurons]` binary fire mask), computes pairwise true Jaccard/IoU (`intersection / union`, not the notebook's original off-by-factor formula — see Section 3), and returns `(twins, dead)`:
 - `twins`: list of `(i, j, jaccard_score)` for pairs with `jaccard >= threshold` and `i < j`, excluding pairs where both neurons never fired.
 - `dead`: neuron indices that never fired at all (jaccard undefined, handled via the `union == 0 → 1` guard to avoid div-by-zero).

__init__.py

Re-exports the full public API — everything a consumer needs is importable directly as `from pruning_transformer import x`. If you add a new public class/function, **add it here too** (both the `from .module import X` line and the `__all__` entry) — nothing in `experiments/` should ever import from a submodule path directly.

6. End-to-end usage patterns

Max-3 saliency pruning (weight-based)

```
from pruning_transformer import Max3SaliencyScorer, SaliencySelector, prune_ffn_layer

selector = SaliencySelector(Max3SaliencyScorer(), prune_percent=40)
kept = prune_ffn_layer(model.bert, layer_index=0, selector=selector)
```

Co-activation twin pruning (behavior-based)

```
from pruning_transformer import ActivationRecorder, JaccardTwinFinder, TwinRedundancySelector, prune_ffn_layer

recorder = ActivationRecorder(model=model.bert, tokenizer=tokenizer)
fires = recorder.record(texts, layer_idx=0)
twins, dead = JaccardTwinFinder().find(fires, threshold=0.95)
kept = prune_ffn_layer(model.bert, layer_index=0, selector=TwinRedundancySelector(twins))
```

Note both flows call the *exact same* `prune_ffn_layer` — the only thing that changes is which `NeuronSelector` is constructed. This is the payoff of the OCP/LSP design: adding a third criterion means writing a third `NeuronSelector` subclass, never touching `prune_ffn_layer` or `FFNSurgeon`.

Scanning a whole network (VGG or BERT)

```
from pruning_transformer import discover_layers, Max3SaliencyScorer, NetworkSaliencyScanner

# VGG: conv layers only
layers = discover_layers(vgg_model, include_conv=True, include_linear=False)
# BERT: FFN-expansion Linears only
layers = discover_layers(bert_model, include_conv=False, include_linear=True, linear_name_filter="intermediate")

scanner = NetworkSaliencyScanner(layers, Max3SaliencyScorer())
report = scanner.scan() # pandas DataFrame
weakest = scanner.weakest_units("features.24", n=5)
```

Attention-head redundancy

```
from pruning_transformer import AttentionHeadAnalyzer

analyzer = AttentionHeadAnalyzer(model) # BertLayerAdapter built automatically
sim = analyzer.compute_similarity(layer_idx=10)
dist = analyzer.compute_distance_matrix(layer_idx=10, metric="manhattan", normalize=True)
```

Targeting a non-BERT model (extension example)

```
from pruning_transformer import FFNLayerAdapter, prune_ffn_layer

class MyModelAdapter(FFNLayerAdapter):
    def get_ffn(self, layer_idx):
        block = my_model.blocks[layer_idx]
        return block.up_proj, block.down_proj

    def set_ffn(self, layer_idx, intermediate, output):
        block = my_model.blocks[layer_idx]
        block.up_proj, block.down_proj = intermediate, output

prune_ffn_layer(my_model, layer_index=0, selector=selector, adapter=MyModelAdapter())
```

No edits to `pruning_workflow.py`, `ffn_surgery.py`, or any selector needed — this is the DIP payoff from Section 4.1.

7. Experiments walkthrough

Each script under `experiments/` is a standalone, runnable exploration stage — narrative order matters (each stage builds conceptually on the last) but there's no code dependency between them; every script only imports from `pruning_transformer`.

#	Script	What it validates	Package pieces exercised
01	<code>01_vgg_saliency_demo.py</code>	Vectorized Max-3 score on one VGG16 conv layer	<code>Max3SaliencyScorer</code> , <code>lowest_scoring</code>
02	<code>02_vgg_network_scan.py</code>	Max-3 stats across all of VGG16	<code>discover_layers</code> , <code>NetworkSaliencyScanner</code>
03	<code>03_bert_universal_scan.py</code>	Max-3 crossed over onto BERT-tiny's FFN layers	same scanner as 02, different <code>discover_layers</code> args
04	<code>04_bert_pruning_surgery_demo.py</code>	Neuron surgery doesn't break a forward pass	<code>SaliencySelector</code> , <code>prune_ffn_layer</code>
05	<code>05_bert_mrpc_full_experiment.py</code>	Real accuracy: baseline → prune (40%) → heal, on MRPC	full Max-3 pipeline + HF <code>Trainer</code>
06	<code>06_attention_head_similarity.py</code>	Cosine similarity between attention heads	<code>AttentionHeadAnalyzer.compute_similarity</code>
07	<code>07_attention_head_distance.py</code>	Manhattan vs. Euclidean head distance	<code>compute_distance_matrix</code> , both metrics
08	<code>08_attention_scaled_distribution.py</code>	Distance on unit-normalized heads (magnitude removed)	<code>compute_distance_matrix(..., normalize=True)</code>
09	<code>09_coactivation_twin_scan.py</code>	Behavior-based twin-neuron detection	<code>ActivationRecorder</code> , <code>JaccardTwinFinder</code>
10	<code>10_twin_neuron_pruning_experiment.py</code>	Real accuracy: prune twins → heal, on MRPC	full twin pipeline + HF <code>Trainer</code>

Stages 05 and 10 are the two "real experiment" scripts — they fine-tune, measure damage, and re-heal. Everything else is a faster diagnostic/demo that doesn't require training.

8. Extension guide — "how do I..."

...add a new saliency scoring rule (e.g. SVD-based)? Write a new class implementing `SaliencyScorer.score(weight) -> Tensor` in `scoring.py` (or a new file, then export it from `__init__.py`). It slots into both `NetworkSaliencyScanner` and `SaliencySelector` with zero other changes.

...add a new neuron-selection criterion? Write a new class implementing `NeuronSelector.select_keep_indices(weight) -> list` in `selectors.py`. It slots into `prune_ffn_layer` as-is.

...prune a new module type (e.g. LayerNorm)? Pass `extra_kinds=[LayerKind("layernorm", nn.LayerNorm)]` to `discover_layers` at the call site — no changes to `layers.py`.

...add a new attention distance metric? Add an entry to `LP_METRICS` in `head_analysis.py` (only works for Lp-norms; a fundamentally different metric, e.g. cosine distance as a *distance* rather than similarity, would need a small refactor of `compute_distance_matrix` beyond the dict lookup).

...support a new model family (e.g. DistilBERT, a custom architecture)? Write a new class implementing `FFNLayerAdapter` and/or `AttentionLayerAdapter` (only implement the one(s) you actually need — see Section 6's example). Pass an instance via the `adapter=` parameter to `prune_ffn_layer`, `AttentionHeadAnalyzer`, or `ActivationRecorder`. Don't touch any of those three files.

...change how FFN surgery physically works? Don't, unless the resize logic itself is wrong. `FFNSurgeon` is deliberately not an interface — there's one correct way to do this operation (see Section 4.2). If you do need to change it, it's the one place selection strategies never see, so the blast radius is contained to `ffn_surgery.py` itself.

9. Lineage: old notebook names → current locations

Useful if you're cross-referencing the original exploratory Colab notebook, older commits, or the thesis framework's naming:

Old / notebook name	Current location
<code>compute_saliency_score_channel_vectorized</code>	<code>scoring.Max3SaliencyScorer</code> + <code>scoring.lowest_scoring</code>
SaliencyPruner (VGG-only)	<code>network_scanner.NetworkSaliencyScanner</code> (generalized)
UniversalPruner (VGG+BERT)	<code>network_scanner.NetworkSaliencyScanner</code> (same class, different <code>discover_layers</code> args)
PruningSurgeon	<code>ffn_surgery.FFNSurgeon</code> + <code>pruning_workflow.prune_ffn_layer</code>
SimilarityScanner	<code>head_analysis.AttentionHeadAnalyzer.compute_similarity</code>
DistributionScanner	<code>head_analysis.AttentionHeadAnalyzer.compute_distance_matrix</code>
CoActivationScanner	split into <code>activation_recording.ActivationRecorder</code> (recording) + <code>redundancy.JaccardTwinFinder</code> (analysis)
TwinSurgeon	<code>selectors.TwinRedundancySelector</code> + <code>pruning_workflow.prune_ffn_layer</code>
<code>attention_similarity.py</code> (early module name)	<code>head_analysis.py</code>
<code>co_activation.py</code> (early module name)	split into <code>activation_recording.py</code> + <code>redundancy.py</code>
<code>surgeon.py</code> (early module name)	<code>ffn_surgery.py</code>
<code>cnn_saliency.py</code> / <code>universal_saliency.py</code> (early module names)	merged into <code>scoring.py</code> + <code>network_scanner.py</code>

10. Known issues / gotchas checklist

- **Nothing has been run end-to-end in this repo layout yet** (see Section 3). Before citing any accuracy number, actually run the corresponding `experiments/05` or `experiments/10` script.
- **No GPU/torch locally** — don't try to `import torch` or run experiments on the primary dev machine; use Colab/Kaggle.
- **FFNSurgeon.resize has no index validation** — an invalid `keep_indices` list will surface as a raw tensor-indexing error, not a friendly message. If you see a confusing `IndexError` from surgery, check the selector that produced the indices first.
- **ActivationRecorder is not reentrant** — don't call `.record()` concurrently on the same instance from multiple threads; sequential reuse (same instance, different `layer_idx`, one call at a time) is fine.
- **TwinRedundancySelector ignores the neuron weights** it's handed — this is correct behavior (the signal is behavioral), not a bug, but it can look surprising in a debugger.
- **BertLayerAdapter also covers RoBERTa** (same `encoder.layer[i]` layout) but nothing else — don't assume it works for GPT-style or T5-style models without checking the actual attribute layout first.

11. Where to go next

- Read `README.md` for the shorter/marketing-level architecture summary.
- Read the module docstrings — every file in `pruning_transformer/` has a top-of-file docstring explaining *why* it's shaped the way it is, written for exactly this kind of handoff.
- Run `experiments/04` first if you want the fastest possible sanity check that the whole pruning pipeline works end-to-end (no training required, just a forward-pass check).